Subhransu S. Bhattacharjee

School of Computing, The Australian National University, Australia

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1ssb • 1ssb • Google Scholar

PhD candidate at ANU, researching multimodal representations for tackling uncertainty in robotic perception. Open to Applied Scientist, Quantitative Researcher, Research Scientist, and Machine Learning Researcher roles across quantitiative trading, generative AI, computer vision and LLM/RAG systems. Builds uncertainty-aware world models and production-facing ML pipelines, published in top-tier conferences.

Education

Doctor of Philosophy — **Artificial Intelligence**
School of Computing, Australian National University, Australia — *Apr 2023 – Present*
- **Thesis topic:** *Into the Unknown: Tackling Spatio-semantic Uncertainty using Generative Posteriors*
- **Supervisors:** Dr. Rahul Shome, Dr. Dylan Campbell, and Prof. Stephen Gould

Bachelor of Engineering — **Mechatronics**
School of Engineering, Australian National University, Australia — *Jul 2018 – Dec 2022*
- **Major:** Mechatronic Systems Engineering, *First Class Honours*
- **Minors:** Mathematics and Electronic Communication Systems; summer school at the London School of Economics
- **Certifications:** Game Theory, Stanford; Machine Learning Production, DeepLearning.AI; Project Management, Google
- **Thesis project:** Whiplash Gradient Descent Dynamics

Internship Experience

Microsoft Corporation — **May 2026 – July 2026**
Applied Scientist PhD Research Intern, LLMs, RAG, Agentic AI
- Worked with the Microsoft India Development Center Science team for LLM-as-judge and RAG pipeline evaluation frameworks for calibration of metrics and groundedness of results from noisy context.

Optiver APAC — **Nov 2024 – Feb 2025**
Quantitative Research Intern, Machine Learning, Market Microstructure
- Built a real-time ML research pipeline for Korean market microstructure, including feature engineering, monitoring, and walk-forward backtesting for rapid alpha strategy evaluation.

JP Morgan Chase & Co. — **Dec 2021 – Feb 2022**
Quantitative Research Intern, NLP, Information Retrieval, Risk
- Built a transformer-based question-answering pipeline over large financial document collections, including document indexing, retrieval, and client-facing research workflow components.

Defence Research & Development Organisation, India — **Apr 2020 – Jul 2020**
Summer Research Trainee, Radar Signal Processing
- Implemented an FPGA-based multiprocessor interface for real-time data exchange and long-range passive radar signal analysis using extended Kalman matched filtering.

Academic Experience

Student Hub, Australian National University — **Feb 2026 – Apr 2026**
Quantitative Researcher for the ANU Student Hub, AI-assisted Optimisation
- Developed an AI-assisted workflow for timetable optimisation and Student Hub usability improvement.

Research School of Management, Australian National University — **Sep 2023 – Sep 2024**
Graduate Research Assistant — FinTech & AI, PI: Dr Priya Muthukannan
- Built assessment frameworks for banks' AI adoption and organisational responses to technological change in an industry-linked project with Commonwealth Bank of Australia.
- Analysed open-banking regimes using a dynamic-capabilities lens to inform strategy recommendations.

CIICADA Lab, Australian National University — **Dec 2022 – Feb 2023**
Summer Research Scholar — Control & Optimisation, Supervisor: Prof Ian Petersen
- Studied feedback-based optimisation methods for systems without closed-form solutions, contributing to publications at ASCC 2022 and ANZCC 2021.

School of Computing, Australian National University — **Mar 2022 – Jun 2022**
Undergraduate Research Scholar — Geometry of Data, Supervisor: Prof Richard Hartley
- Investigated invertibility of differentiable neural mappings using neural networks; achieved a 72% hit rate under an RMSE criterion using dense positional encoding.

Selected Publications

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *Believing is Seeing: Unobserved Object Detection using Generative Models. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Project page.
Uses pre-trained generative models to infer spatio-semantic distributions of occluded and out-of-frame objects across 2D, 2.5D, and 3D scenes.

Subhransu S. Bhattacharjee, Dylan Campbell & Rahul Shome: *FlatLands: Generative Floormap Completion From a Single Egocentric View. European Conference on Computer Vision (ECCV)*, 2026. Springer. Paper.
Releases FlatLands: 270,575 observations from 17,656 real metric indoor scenes across six datasets.
Compares 11 methods, showing stochastic generators improve completion fidelity and capture layout uncertainty.

Subhransu S. Bhattacharjee, Hao Lu, Dylan Campbell & Rahul Shome: *MatterDoor: Sampling Zero-shot Spatio-semantic Priors using Generative Models. arXiv:2510.11014*, 2025. Under review; RSS 2026 workshop presentation, Sydney. Preprint.
Samples environment priors from partial observations using pretrained generative models.
Produces RGB-D point-cloud hypotheses for occupancy, target semantics, and configuration-space planning.

Subhransu S. Bhattacharjee & Ian Petersen: *Analysis of the Whiplash Gradient Descent Dynamics. Asian Journal of Control* (Special Issue), Wiley, 2023. DOI: 10.1002/asjc.3153.
A feedback-based inertial gradient method with adaptive damping achieves provable polynomial and exponential convergence rates that extend classical optimisation theory to high-dimensional settings.

Subhransu S. Bhattacharjee & Ian Petersen: *A Closed Loop Gradient Descent Algorithm Applied to Rosenbrock's Function. Australian & New Zealand Control Conference (ANZCC)*, 2021. DOI: 10.1109/ANZCC53563.2021.9628258.
Introduces adaptive closed-loop damping for inertial gradient descent on a stiff non-convex optimisation benchmark.

Achievements

2026 Google Research, Granted with ANU Robotics $76,000 for the Development of World Models for Task and Motion Planning
2026 Invited to present the FlatLands paper at the Hugging Face, PyTorch, and Red Hat Meetup, Bengaluru.
2025 Invited Research Talk at the Dyson Robotics Laboratory, Imperial College London
2025 Vice-Chancellor's Travel Grant, Australian National University, for CVPR 2025 travel.
2025 Invited attendee, Citadel & Citadel Securities PhD Summit, London.
2025 Invited attendee, Citadel Securities Australia, Summer Networking Event, Sydney.
2024 Optiver PhD Quant Lab Program (selected participant).
2023 ANU International University Research Scholarship (4 years) and Higher Degree by Research Merit Stipend.
2022 Highly Recommended Paper at the *Asian Control Conference*; Best Student Reviewer Award.
2022 Course highest in ENGN4627: Robotics.
2021 High Commendation Award for undergraduate student paper at the *Australia & New Zealand Control Conference*.
2020 ANU Chancellor's International Scholarship for academic merit for transfer students.
2019 Chancellor's Achiever and Special Achiever Award for Academic Merit

Teaching & Service

Teaching

- **Senior Tutor, Introduction to Machine Learning; Professional Practices: Responsible Leadership** — School of Computing, ANU (2025 – 2026): Taught machine-learning mathematics and optimisation; ran tutorials, marked exams, and assessed projects.
- **Head Tutor, Building Cyber-Physical Systems** — School of Cybernetics, ANU (Mar 2025 – Nov 2025): Co-designed and delivered project-based modules in microprocessors, robotics, and machine learning; led NAO robot labs; coordinated assessments.
- **Tutor, Power Systems & Control Theory** — School of Engineering, ANU (Jul 2022 – Sep 2023): Ran labs for ENGN8824, workshops for ENGN4628, and targeted tutoring for ENGN4625.

Service

Reviewing: ECCV 2026; NeurIPS 2026; ICML 2026; CVPR 2026; ICRA 2025–2026; IEEE Robotics and Automation Letters (RA-L) 2026; AAAI 2025; IROS 2025; Asian Journal of Control (2023); American Control Conference (2022); ANZCC (2021–2022).
Volunteering: Friends of Tribal Society volunteer (2017); Set4ANU HDR mentor (2024); TechLauncher manager (2024).

Technical Skills

Programming & ML Frameworks: Python; PyTorch; SQL; C; MATLAB; Bash; NumPy; pandas; SciPy; scikit-learn; PySpark; LaTeX.
CV, Robotics & Generative AI: OpenCV; Open3D; PyTorch3D; TorchVision; ROS2; COLMAP; 3D reconstruction; world models; diffusion models; flow matching; uncertainty estimation.
LLMs, NLP & Retrieval: transformer models; supervised fine-tuning; RLHF; RLAIF; DPO; RAG; vector databases; information retrieval; document indexing; agentic tool selection.
Finance & Quant Research: market microstructure; alpha research; time-series forecasting; risk modelling; backtesting.
Systems, Data & Deployment: Linux; Git; Docker; Apptainer; SLURM; CUDA programming; Numba; CuPy; `torch.compile`; ONNX Runtime; Weights & Biases; AWS — EC2, S3; REST APIs; backtesting frameworks; AMD Vivado.